

1. Equilibrium and Elasticity

1.1 Equilibrium

Conditions for Equilibrium

1. $F^{\text{ext}} = 0$
2. $\tau^{\text{ext}} = 0$

1.2 Strain, Stress, and Elastic Modulus

Just remember some definitions

Some definition:

$$\text{stress} = \frac{\text{force}}{\text{area}}$$

$$\text{strain} = \frac{\Delta l (\text{deformation})}{l_0}$$

$$\text{elastic modulus} = \frac{\text{stress}}{\text{strain}}$$

$$\text{tensile stress} = \frac{F_{\perp}}{A}$$

$$\text{tensile strain} = \frac{\Delta l}{l_0}$$

Young's modulus is tensile stress divided by tensile strain, and is given by

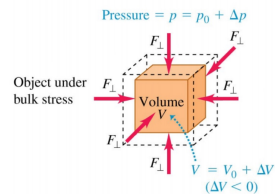
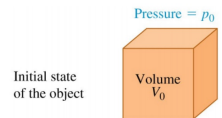
$$Y = \frac{\frac{F_{\perp}}{A}}{\frac{\Delta l}{l_0}}$$

bulk stress and strain

$$\text{pressure } P = \frac{F_{\perp}}{A}$$

$$\text{bulk stress} : \Delta p$$

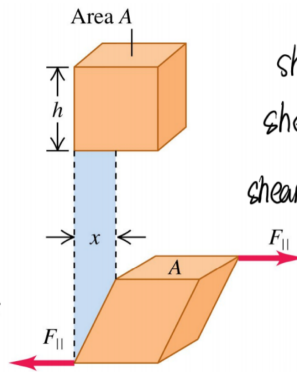
$$\text{bulk strain} : \frac{\Delta V}{V_0}$$

$$\text{bulk modulus } B = - \frac{\Delta P}{\frac{\Delta V}{V_0}} \quad (\Delta V < 0)$$


shear stress & strain

Initial state of the object

Object under shear stress



shear stress $\frac{F_{\parallel}}{A}$

shear strain $\frac{x}{h}$

shear modulus $S = \frac{\frac{F_{\parallel}}{A}}{\frac{x}{h}}$

2. Fluid Dynamics

2.1 Some Concepts

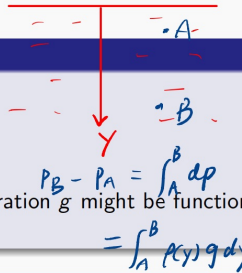
Pressure in Liquid

Equations

$$dp = \rho g dy$$

$$\Delta p = \int_{p_1}^{p_2} dp$$

*The density of mass ρ and the gravity acceleration g might be function of depth y .



Pressure with Constant ρ and g

$$p = p_0 + \rho gh$$

atmospheric pressure: $1.01 \times 10^5 \approx 1 \times 10^5 [Pa]$

pressure in 1-meter-depth water: $9.8 \times 10^3 \approx 1 \times 10^4 [Pa]$

*U-shaped pipe



← surfaces w/ same height (still)

Buoyant Force

$$F_b = \rho_{\text{liquid}} V_{\text{immersed}} g$$

The buoyant force equals to the weight of the fluid displaced by the body in magnitude.

Practically, the density and the gravity acceleration can still be function of depth y . In this case, solve the buoyant force with integration.

*For objects with vertical side surfaces, buoyant force can be derived with $F_b = (p_2 - p_1)A$.

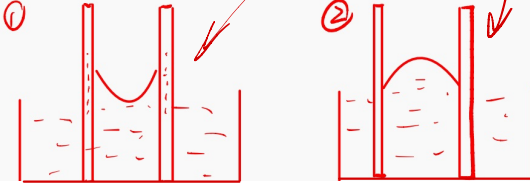
Surface Tension

Origin

The attractive interaction between molecules of the liquid.

Meniscus and capillarity

- ① adhesion forces dominate – concave liquid surface
- ② cohesion forces dominate – convex liquid surface



★ $p + \frac{1}{2}\rho v^2 + \rho g y = \text{const}$ → Bernoulli's Equation

1.2 Exercises

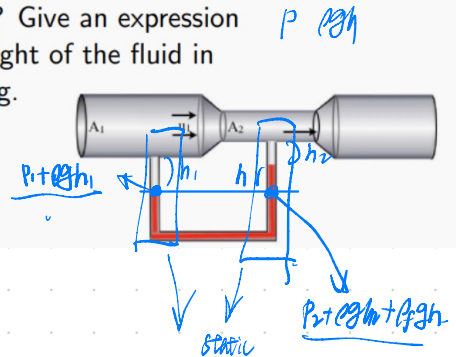
The wide section of a pipe (with cross-sectional area A_1) in the apparatus shown below transports an incompressible fluid of density ρ with an unknown flow speed. The wide and narrow section (of cross-sectional area A_2) of the pipe are connected via a U-shaped pipe that is filled with another incompressible fluid (marked in red) of density $\rho_f > \rho$. How can this apparatus can be used to determine the flow speed in the wider section of the tube? Give an expression for the speed in terms of A_1 , A_2 , ρ , ρ_f , the height of the fluid in the U-pipe and the acceleration due to gravity g .

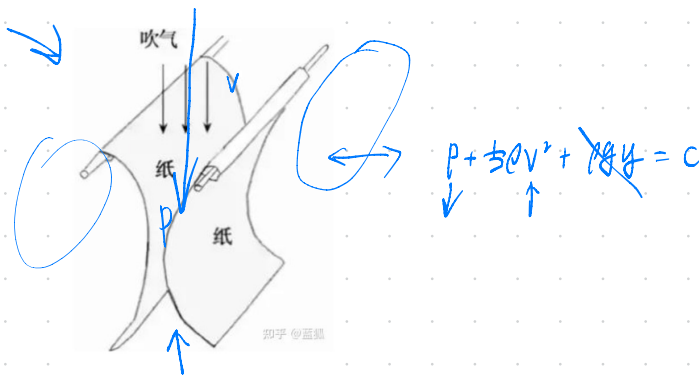
① $A_1 \cdot v_1 = A_2 \cdot v_2$

② $p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$

③ $p_1 + \rho_f g h_1 = p_2 + \rho_f g h_2 + \rho_f g h$

⇒ $v_1 = \sqrt{\frac{2(\rho_f - \rho)gh}{\rho \left(\left(\frac{A_1}{A_2} \right)^2 - 1 \right)}}$





A vertical cylindrical container has a height h and a radius R . The top is open, and the bottom has a small hole with an area S . ($S \ll \pi R^2$)

- (1) Find the outflow velocity of water through the small hole at the bottom.
- (2) Find the time it takes for the container to completely empty.

(1) $\rho gh + \frac{1}{2} \rho v_0^2 + \underbrace{P}_{P_{atm}} = \frac{1}{2} \rho v_0^2 + \underbrace{P}_{P_{atm}}$

$v_0 \approx \sqrt{2gh}$

(2) $S_0 v_A = S v_0$
 $v_A = -\frac{dh}{dt}$ $v_0 = \sqrt{2gh}$

$\Rightarrow -\frac{S}{2S_0} \sqrt{2g} dt = d(\sqrt{h})$

$\int_0^t -\frac{S}{2S_0} \sqrt{2g} dt = \int_h^0 d(\sqrt{h})$

$\Rightarrow t = \frac{\pi R^2}{S} \sqrt{\frac{2h}{g}}$

As shown in the figure, the lower outlet of container **A** is connected to a horizontal pipe. Point **c** in the middle of the pipe is a **constriction** (narrow section), and the **d** end of the horizontal pipe is open to the atmosphere.

(1) Find the pressure at point **c**.

(2) Under what condition can the liquid in container **B** be drawn into the horizontal pipe due to suction (i.e., when will the pipe create suction that causes inflow from B)?

$$(1) Vd \approx \sqrt{2gh}$$

$$S_c V_c = S_d V_d$$

$$P_0 + \frac{1}{2} \rho V_d^2 = P_c + \frac{1}{2} \rho V_c^2$$

$$\Rightarrow P_c = P_0 + \rho gh \left[1 - \left(\frac{S_d}{S_c} \right)^2 \right]$$

(2)

$$P_c < P_0$$

$$P_E = P_c + \rho gh_b$$

$$P_E - P_0 = \rho gh \left[1 - \left(\frac{S_d}{S_c} \right)^2 \right] + \rho gh_b < 0$$

$$\Rightarrow \frac{S_d}{S_c} > \sqrt{1 + \frac{h_b}{h}}$$

